

Association of Age and Glioblastoma - GDC Data

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```
library(tidyverse)
library(janitor)
library(knitr)
library(broom)

options(scipen = 999)
theme_set(theme_minimal(base_size = 12))
```

Introduction

I used open-access clinical data downloaded from the National Cancer Institute GDC Data Portal. The dataset used here is from the Foundation Medicine Adult Cancer Clinical Dataset (FM-AD), of spinal cord, cranial nerves, and other central nervous system tumor cases. This code includes basic data import, data quality checks, cohort characterization, visualization, and regression modeling. The analysis is only intended for demonstration of coding and analytic approach.

1. Loading Data

This section reads the GDC clinical TSV file. If the downloaded .tar.gz file is available in the working directory, the code extracts it.

```
untar("gdc_download_20260624_005957.719701.tar.gz", exdir = "gdc_data")
list.files("gdc_data", recursive = TRUE)
```

```
## [1] "939753cd-bcf0-4d48-b8f1-1d0dea65bd69/FM-AD_Clinical.Spinal_Cord_Cranial_Nerves_And_Other_Parts_..."
## [2] "MANIFEST.txt"
```

```
library(readr)

clinical_raw <- read_tsv(
  "gdc_data/939753cd-bcf0-4d48-b8f1-1d0dea65bd69/FM-AD_Clinical.Spinal_Cord_Cranial_Nerves_And_Other_Pa..."
)
```

```
## Rows: 971 Columns: 17
## -- Column specification -----
## Delimiter: "\t"
## chr (16): cases.submitter_id, case_id, project_id, project.name, cases.disea...
```

```
## dbl (1): diagnoses.age_at_diagnosis
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
clinical_data <- clinical_raw %>%
  clean_names()

glimpse(clinical_data)
```

```
## Rows: 971
## Columns: 17
## $ cases_submitter_id      <chr> "AD10063", "AD10185", "AD10194", ~
## $ case_id                 <chr> "9ecfffaf-1eb3-4189-93f2-2ef6423~
## $ project_id              <chr> "FM-AD", "FM-AD", "FM-AD", "FM-A~
## $ project_name             <chr> "Foundation Medicine Adult Cance~
## $ cases_disease_type      <chr> "Gliomas", "Gliomas", "Gliomas", ~
## $ cases_primary_site      <chr> "Spinal Cord, Cranial Nerves, An~
## $ demographic_submitter_id <chr> "AD10063_demographic", "AD10185_~
## $ demographic_id          <chr> "fc4bf536-1824-4f62-9182-5c5641e~
## $ demographic_gender      <chr> "male", "female", "male", "male"~
## $ diagnoses_submitter_id  <chr> "AD10063_diagnosis", "AD10185_di~
## $ diagnosis_id            <chr> "13a4b838-feec-4428-bb53-e01cf17~
## $ diagnoses_primary_diagnosis <chr> "Astrocytoma, NOS", "Glioblastom~
## $ diagnoses_site_of_resection_or_biopsy <chr> "Brain, NOS", "Brain, NOS", "Bra~
## $ diagnoses_classification_of_tumor <chr> "metastasis", "metastasis", "met~
## $ diagnoses_tissue_or_organ_of_origin <chr> "Nervous system, NOS", "Nervous ~
## $ diagnoses_age_at_diagnosis <dbl> 21367, 20050, 17944, 14700, 2080~
## $ diagnoses_morphology    <chr> "9400/3", "9440/3", "9450/3", "9~
```

2. Data Quality

```
data_quality <- clinical_data %>%
  summarise(
    total_rows = n(),
    unique_cases = n_distinct(case_id),
    unique_submitter_ids = n_distinct(cases_submitter_id),
    missing_gender = sum(is.na(demographic_gender)),
    missing_age_at_diagnosis = sum(is.na(diagnoses_age_at_diagnosis)),
    missing_primary_diagnosis = sum(is.na(diagnoses_primary_diagnosis)),
    missing_classification = sum(is.na(diagnoses_classification_of_tumor)),
    missing_site = sum(is.na(diagnoses_site_of_resection_or_biopsy))
  )

kable(data_quality, digits = 1, caption = "Data Quality Summary")
```

Table 1: Data Quality Summary

total_rows	unique_cases	unique_submitter_id	missing_gender	missing_age_at_diagnosis	missing_primary_site	missing_classification	missing_site
971	971	971	0	0	0	0	0

```
missingness <- clinical_data %>%
  summarise(across(everything(), ~ sum(is.na(.)))) %>%
  pivot_longer(
    cols = everything(),
    names_to = "variable",
    values_to = "missing_count"
  ) %>%
  mutate(
    missing_percent = missing_count / nrow(clinical_data) * 100
  ) %>%
  arrange(desc(missing_percent))

kable(missingness, digits = 1, caption = "Missingness by Variable")
```

Table 2: Missingness by Variable

variable	missing_count	missing_percent
cases_submitter_id	0	0
case_id	0	0
project_id	0	0
project_name	0	0
cases_disease_type	0	0
cases_primary_site	0	0
demographic_submitter_id	0	0
demographic_id	0	0
demographic_gender	0	0
diagnoses_submitter_id	0	0
diagnosis_id	0	0
diagnoses_primary_diagnosis	0	0
diagnoses_site_of_resection_or_biopsy	0	0
diagnoses_classification_of_tumor	0	0
diagnoses_tissue_or_organ_of_origin	0	0
diagnoses_age_at_diagnosis	0	0
diagnoses_morphology	0	0

Age at diagnosis is stored in days in the GDC file, so it is converted to years.

```
analytic_data <- clinical_data %>%
  mutate(
    age_at_diagnosis_yrs = diagnoses_age_at_diagnosis / 365.25,
    age_group = case_when(
      age_at_diagnosis_yrs < 40 ~ "Under 40",
      age_at_diagnosis_yrs >= 40 & age_at_diagnosis_yrs < 65 ~ "40-64",
      age_at_diagnosis_yrs >= 65 ~ "65+",
      TRUE ~ NA_character_
    )
  )
```

```

),
diagnosis_group = case_when(
  str_detect(str_to_lower(diagnoses_primary_diagnosis), "glioblastoma") ~ "Glioblastoma",
  str_detect(str_to_lower(diagnoses_primary_diagnosis), "astrocytoma") ~ "Astrocytoma",
  str_detect(str_to_lower(diagnoses_primary_diagnosis), "oligodendroglioma") ~ "Oligodendroglioma",
  str_detect(str_to_lower(diagnoses_primary_diagnosis), "meningioma") ~ "Meningioma",
  str_detect(str_to_lower(diagnoses_primary_diagnosis), "glioma") ~ "Other glioma",
  TRUE ~ "Other CNS tumor"
),
glioblastoma_flag = if_else(diagnosis_group == "Glioblastoma", 1, 0),
demographic_gender = str_to_title(demographic_gender),
age_group = factor(age_group, levels = c("Under 40", "40-64", "65+")),
diagnosis_group = factor(diagnosis_group),
demographic_gender = factor(demographic_gender)
)

analytic_data %>%
  select(
    cases_submitter_id,
    demographic_gender,
    age_at_diagnosis_yrs,
    age_group,
    diagnoses_primary_diagnosis,
    diagnosis_group,
    diagnoses_classification_of_tumor,
    diagnoses_site_of_resection_or_biopsy
  ) %>%
  head(10) %>%
  kable(digits = 1, caption = "First 10 Analytic Records")

```

Table 3: First 10 Analytic Records

cases_submitter_id	demographic_gender	age_at_diagnosis_yrs	age_group	diagnoses_primary_diagnosis	diagnosis_group	diagnoses_classification_of_tumor	diagnoses_site_of_resection_or_biopsy
AD10063	Male	58.5	40-64	Astrocytoma, NOS	Astrocytoma	metastasis	Brain, NOS
AD10185	Female	54.9	40-64	Glioblastoma, NOS	Glioblastoma	metastasis	Brain, NOS
AD10194	Male	49.1	40-64	Oligodendroglioma, NOS	Oligodendroglioma	metastasis	Brain, NOS
AD10229	Male	40.2	40-64	Glioblastoma, NOS	Glioblastoma	metastasis	Brain, NOS
AD10231	Male	57.0	40-64	Glioblastoma, NOS	Glioblastoma	metastasis	Brain, NOS
AD10237	Male	47.1	40-64	Glioblastoma, NOS	Glioblastoma	metastasis	Brain, NOS
AD10240	Male	51.7	40-64	Glioblastoma, NOS	Glioblastoma	metastasis	Brain, NOS
AD10249	Female	23.0	Under 40	Astrocytoma, NOS	Astrocytoma	metastasis	Brain, NOS
AD10251	Female	69.9	65+	Glioblastoma, NOS	Glioblastoma	metastasis	Brain, NOS

cases_submitter_id	demographic_gender	age_at_diagnosis_yrs	diagnosis_group	diagnoses_primary	diagnoses_secondary	diagnoses_classification	diagnoses_tumor	time_of_resection_or_biopsy
AD10255	Female	32.2	Under 40	Oligodendroglioma NOS	Oligodendroglioma NOS	Brain, NOS		

4. Cohort Summary

```
cohort_summary <- analytic_data %>%
  summarise(
    cases = n_distinct(case_id),
    mean_age = mean(age_at_diagnosis_yrs, na.rm = TRUE),
    median_age = median(age_at_diagnosis_yrs, na.rm = TRUE),
    min_age = min(age_at_diagnosis_yrs, na.rm = TRUE),
    max_age = max(age_at_diagnosis_yrs, na.rm = TRUE),
    female_cases = sum(demographic_gender == "Female", na.rm = TRUE),
    male_cases = sum(demographic_gender == "Male", na.rm = TRUE)
  )

kable(cohort_summary, digits = 1, caption = "Overall Cohort Summary")
```

Table 4: Overall Cohort Summary

cases	mean_age	median_age	min_age	max_age	female_cases	male_cases
971	50.9	52.6	19.2	86.6	381	590

5. Table 1: Characteristics by Diagnosis Group

```
table1 <- analytic_data %>%
  group_by(diagnosis_group) %>%
  summarise(
    n = n(),
    mean_age = mean(age_at_diagnosis_yrs, na.rm = TRUE),
    median_age = median(age_at_diagnosis_yrs, na.rm = TRUE),
    percent_female = mean(demographic_gender == "Female", na.rm = TRUE) * 100,
    percent_male = mean(demographic_gender == "Male", na.rm = TRUE) * 100,
    .groups = "drop"
  ) %>%
  arrange(desc(n))

kable(table1, digits = 1, caption = "Cohort Characteristics by Diagnosis Group")
```

Table 5: Cohort Characteristics by Diagnosis Group

diagnosis_group	n	mean_age	median_age	percent_female	percent_male
Glioblastoma	560	55.0	56.6	38.0	62.0

diagnosis_group	n	mean_age	median_age	percent_female	percent_male
Astrocytoma	188	43.7	42.5	43.1	56.9
Oligodendroglioma	76	46.1	46.5	36.8	63.2
Other glioma	60	45.0	45.3	41.7	58.3
Meningioma	49	54.8	58.4	38.8	61.2
Other CNS tumor	38	39.9	42.2	39.5	60.5

6. Diagnosis Distribution

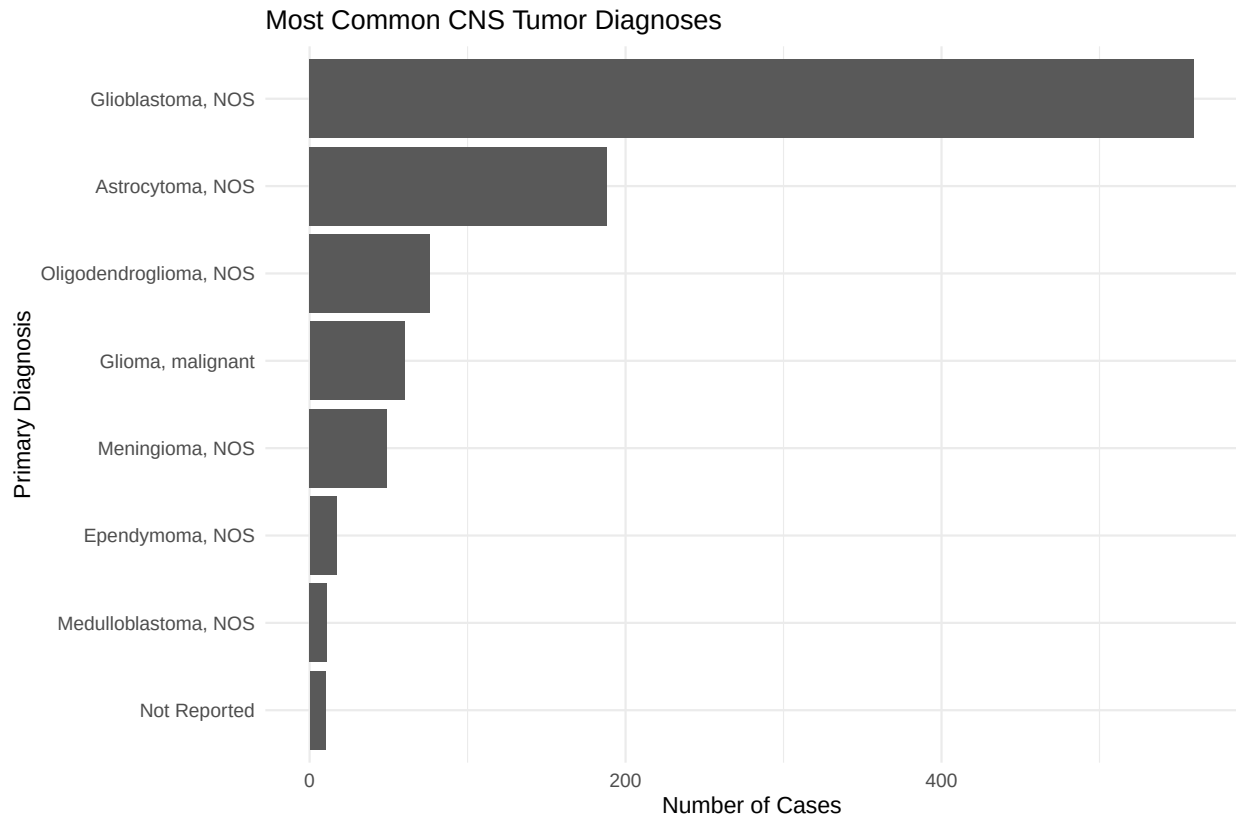
```
diagnosis_counts <- analytic_data %>%
  count(diagnoses_primary_diagnosis, sort = TRUE) %>%
  mutate(percent = n / sum(n) * 100)

kable(head(diagnosis_counts, 15), digits = 1, caption = "Top Primary Diagnoses")
```

Table 6: Top Primary Diagnoses

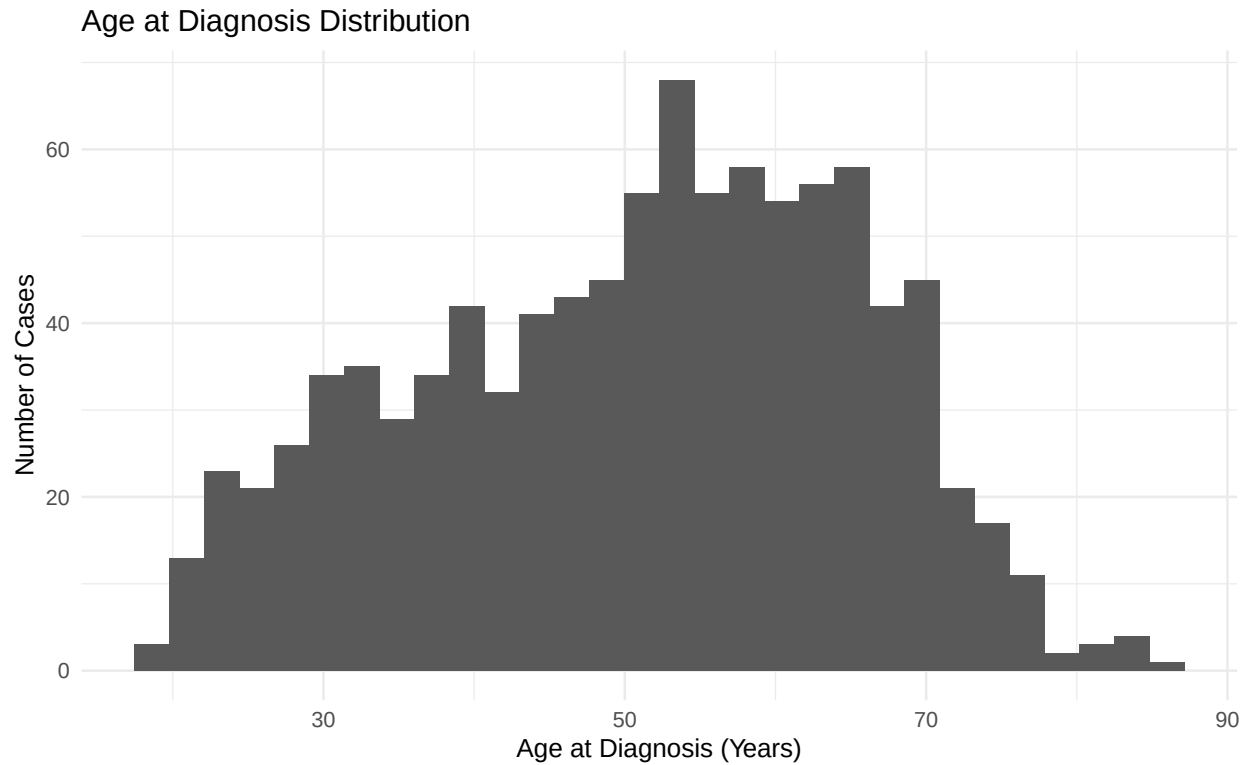
diagnoses_primary_diagnosis	n	percent
Glioblastoma, NOS	560	57.7
Astrocytoma, NOS	188	19.4
Oligodendroglioma, NOS	76	7.8
Glioma, malignant	60	6.2
Meningioma, NOS	49	5.0
Ependymoma, NOS	17	1.8
Medulloblastoma, NOS	11	1.1
Not Reported	10	1.0

```
diagnosis_counts %>%
  slice_max(n, n = 10) %>%
  ggplot(aes(x = reorder(diagnoses_primary_diagnosis, n), y = n)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Most Common CNS Tumor Diagnoses",
    x = "Primary Diagnosis",
    y = "Number of Cases"
  )
```

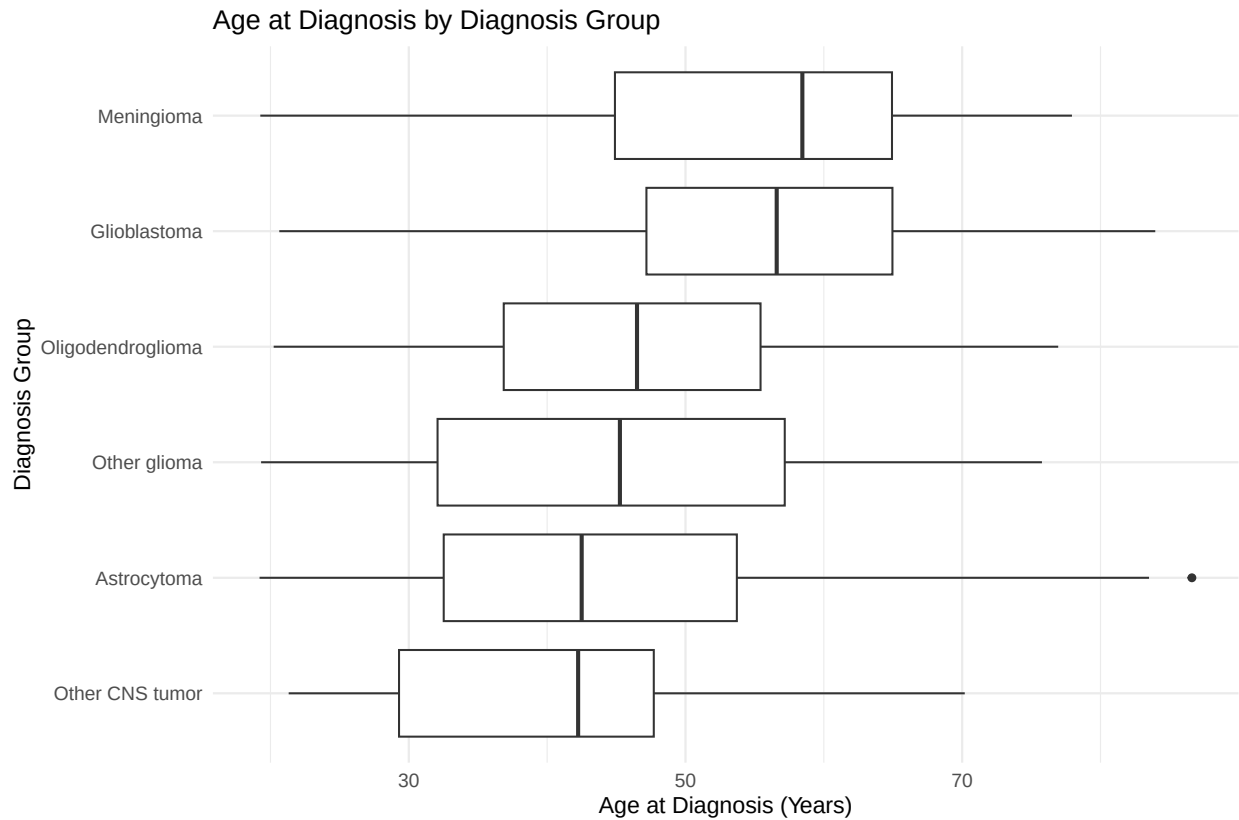


7. Age Distribution

```
ggplot(analytic_data, aes(x = age_at_diagnosis_yrs)) +  
  geom_histogram(bins = 30) +  
  labs(  
    title = "Age at Diagnosis Distribution",  
    x = "Age at Diagnosis (Years)",  
    y = "Number of Cases"  
  )
```



```
ggplot(analytic_data, aes(x = reorder(diagnosis_group, age_at_diagnosis_yrs, median),  
                          y = age_at_diagnosis_yrs)) +  
  geom_boxplot() +  
  coord_flip() +  
  labs(  
    title = "Age at Diagnosis by Diagnosis Group",  
    x = "Diagnosis Group",  
    y = "Age at Diagnosis (Years)"  
  )
```

8. Sex Distribution by Diagnosis Group

```
sex_by_diagnosis <- analytic_data %>%
  count(diagnosis_group, demographic_gender) %>%
  group_by(diagnosis_group) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ungroup()

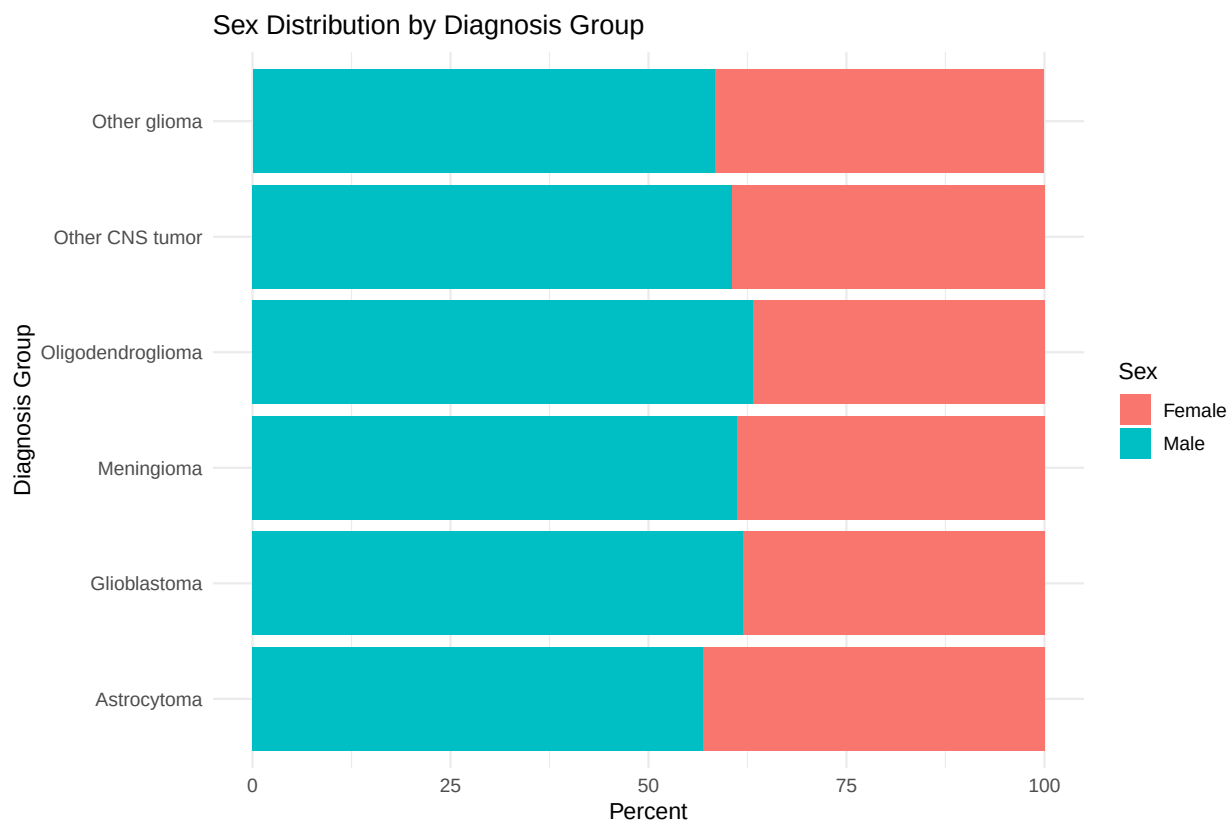
kable(sex_by_diagnosis, digits = 1, caption = "Sex Distribution by Diagnosis Group")
```

Table 7: Sex Distribution by Diagnosis Group

diagnosis_group	demographic_gender	n	percent
Astrocytoma	Female	81	43.1
Astrocytoma	Male	107	56.9
Glioblastoma	Female	213	38.0
Glioblastoma	Male	347	62.0
Meningioma	Female	19	38.8
Meningioma	Male	30	61.2
Oligodendroglioma	Female	28	36.8
Oligodendroglioma	Male	48	63.2
Other CNS tumor	Female	15	39.5
Other CNS tumor	Male	23	60.5

diagnosis_group	demographic_gender	n	percent
Other glioma	Female	25	41.7
Other glioma	Male	35	58.3

```
ggplot(sex_by_diagnosis,
       aes(x = diagnosis_group, y = percent, fill = demographic_gender)) +
  geom_col(position = "stack") +
  coord_flip() +
  labs(
    title = "Sex Distribution by Diagnosis Group",
    x = "Diagnosis Group",
    y = "Percent",
    fill = "Sex"
  )
)
```



9. Age by Diagnosis Group

This section uses a Kruskal-Wallis test to compare age distributions.

```
age_test <- kruskal.test(age_at_diagnosis_yrs ~ diagnosis_group, data = analytic_data)
age_test
```

```
##
```

```
## Kruskal-Wallis rank sum test
##
## data: age_at_diagnosis_yrs by diagnosis_group
## Kruskal-Wallis chi-squared = 137.4, df = 5, p-value <
## 0.000000000000000022
```

10. Sex by Diagnosis Group

```
sex_table <- table(analytic_data$diagnosis_group, analytic_data$demographic_gender)

sex_test <- chisq.test(sex_table)
sex_test
```

```
##
## Pearson's Chi-squared test
##
## data: sex_table
## X-squared = 1.8433, df = 5, p-value = 0.8704
```

11. Factors Associated with Glioblastoma Diagnosis

This model evaluates whether age and sex are associated with having glioblastoma compared with other CNS tumor diagnoses.

```
gbm_model <- glm(
  glioblastoma_flag ~ age_at_diagnosis_yrs + demographic_gender,
  data = analytic_data,
  family = binomial()
)

summary(gbm_model)

##
## Call:
## glm(formula = glioblastoma_flag ~ age_at_diagnosis_yrs + demographic_gender,
##      family = binomial(), data = analytic_data)
##
## Coefficients:
##              Estimate Std. Error z value      Pr(>|z|)
## (Intercept)   -2.231486   0.271225  -8.227 <0.0000000000000002 ***
## age_at_diagnosis_yrs    0.049905   0.005057   9.869 <0.0000000000000002 ***
## demographic_genderMale  0.053581   0.141030   0.380      0.704
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1323.1  on 970  degrees of freedom
## Residual deviance: 1212.5  on 968  degrees of freedom
```

```
## AIC: 1218.5
##
## Number of Fisher Scoring iterations: 4

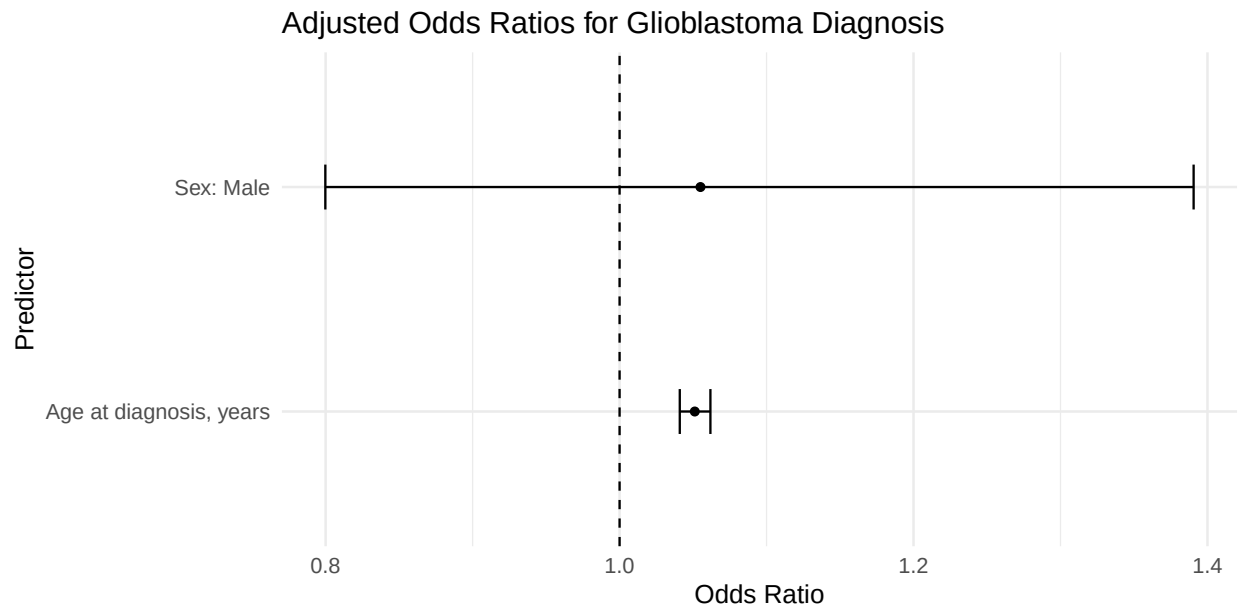
gbm_results <- tidy(gbm_model, conf.int = TRUE, exponentiate = TRUE) %>%
  mutate(
    term = case_when(
      term == "(Intercept)" ~ "Intercept",
      term == "age_at_diagnosis_yrs" ~ "Age at diagnosis, years",
      str_detect(term, "demographic_gender") ~ str_replace(term, "demographic_gender", "Sex: "),
      TRUE ~ term
    )
  )

kable(gbm_results, digits = 3, caption = "Adjusted Odds Ratios for Glioblastoma Diagnosis")
```

Table 8: Adjusted Odds Ratios for Glioblastoma Diagnosis

term	estimate	std.error	statistic	p.value	conf.low	conf.high
Intercept	0.107	0.271	-8.227	0.000	0.063	0.181
Age at diagnosis, years	1.051	0.005	9.869	0.000	1.041	1.062
Sex: Male	1.055	0.141	0.380	0.704	0.800	1.391

```
gbm_results %>%
  filter(term != "Intercept") %>%
  ggplot(aes(x = reorder(term, estimate), y = estimate)) +
  geom_point() +
  geom_errorbar(aes(ymin = conf.low, ymax = conf.high), width = 0.2) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  coord_flip() +
  labs(
    title = "Adjusted Odds Ratios for Glioblastoma Diagnosis",
    x = "Predictor",
    y = "Odds Ratio"
  )
```



12. Interpretation

971 adult central nervous system (CNS) tumor cases were obtained from the National Cancer Institute Genomic Data Commons data. Glioblastoma was the most common diagnosis, accounting for 57.7% of cases, followed by astrocytoma (19.4%) and oligodendroglioma (7.8%). Patients with glioblastoma were older on average and age distributions differed significantly across diagnosis groups based on Kruskal-Wallis test ($p < 0.001$). The proportion of males and females did not differ significantly across diagnosis groups. In the multivariable logistic regression model, increasing age at diagnosis was associated with higher odds of glioblastoma compared to other CNS tumors ($OR = 1.05$ per year increase, 95% CI: 1.04–1.06, $p < 0.001$). Sex was not significantly associated with glioblastoma diagnosis. These findings suggest that age is an important demographic characteristic associated with glioblastoma within this cohort.

```
sessionInfo()
```

```
## R version 4.3.1 (2023-06-16 ucrt)
## Platform: x86_64-w64-mingw32/x64 (64-bit)
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##
## locale:
## [1] LC_COLLATE=English_United States.utf8
## [2] LC_CTYPE=English_United States.utf8
## [3] LC_MONETARY=English_United States.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United States.utf8
##
## time zone: America/New_York
## tzcode source: internal
##
```

```

## attached base packages:
## [1] stats      graphics  grDevices utils      datasets  methods   base
##
## other attached packages:
## [1] broom_1.0.8      knitr_1.47      janitor_2.2.1    lubridate_1.9.3
## [5] forcats_1.0.0    stringr_1.5.1    dplyr_1.1.4      purrr_1.0.4
## [9] readr_2.1.5      tidyr_1.3.1      tibble_3.2.1     ggplot2_3.5.1
## [13] tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] utf8_1.2.4      generics_0.1.3    stringi_1.8.4     hms_1.1.3
## [5] digest_0.6.36   magrittr_2.0.3    evaluate_0.24.0   grid_4.3.1
## [9] timechange_0.3.0 fastmap_1.2.0     backports_1.5.0   fansi_1.0.6
## [13] scales_1.3.0    cli_3.6.3         rlang_1.1.4       crayon_1.5.3
## [17] bit64_4.0.5     munsell_0.5.1     withr_3.0.0       yaml_2.3.8
## [21] tools_4.3.1     parallel_4.3.1    tzdb_0.4.0        colorspace_2.1-0
## [25] vctrs_0.6.5     R6_2.5.1          lifecycle_1.0.4   snakecase_0.11.1
## [29] bit_4.0.5       MASS_7.3-60.0.1   vroom_1.6.5       pkgconfig_2.0.3
## [33] pillar_1.9.0    gtable_0.3.5      glue_1.7.0        highr_0.11
## [37] xfun_0.49       tidyselect_1.2.1  rstudioapi_0.17.1 farver_2.1.2
## [41] htmltools_0.5.8.1 rmarkdown_2.29    labeling_0.4.3     compiler_4.3.1

```